

More about the rainforest

In an earlier activity, you modeled daily rainfall in an equatorial rainforest with independent random variables having the exponential(λ) distribution. To match the average daily rainfall of 80/365 inches per day, you took λ to be 365/80.

1. According to this model, what is the probability that the total rainfall on any given day is less than or equal to x inches? In other words, what is the distribution function for an exponential(λ) random variable? You can look it up in Durrett.
2. What is the probability that the total rainfall for each of the 7 days in a week is less than or equal to x inches? Calculate the probability of the following event:
(rainfall on Sunday $\leq x$) $\cap$ (rainfall on Monday $\leq x$) $\cap$... $\cap$ (rainfall on Saturday $\leq x$). Use the independence of rainfall on different days to find the probability in terms of the distribution function from part a.
3. You have just computed the distribution function for the random variable Y representing the maximum rainfall for a week. If, for example, during a week, it rained 0.516 inches, 0.201 inches, 0.320 inches, 0.129 inches, 0.277 inches, 0.238 inches, 0.199 inches, then Y would be 0.516 inches. For Y to be less than or equal to x , every day must have x or less inches of rainfall. That is the event whose probability you calculated.

Calculate the density function for Y by taking the derivative of your distribution function.

- Use a graphing calculator or Desmos online to graph the density function. Sketch the graph here, including scales on the axes.
- Roughly where is the “mode” of the distribution of Y ? That is the input with highest density. What are the units of measurement for this number?
- Calculate the median for Y by solving the equation $F(m) = \frac{1}{2}$, where F is the distribution function for Y . What are the units of measurement for this number?